

Chatting with Sheldon in VR: How Stylization and Voice Shape Trust in Celebrity Embodied Conversational Agents?

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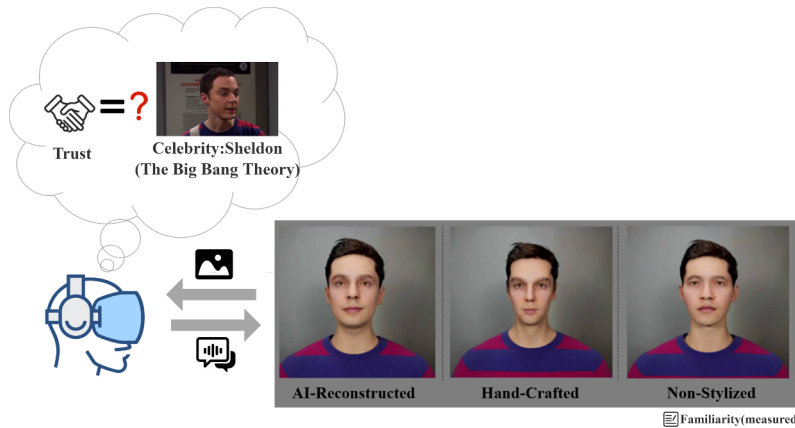


Figure 1: Overview of the study. We examine how *Style* and *Voice* affect users' trust and social presence during conversation. Participants completed six within-subject interaction conditions with a celebrity ECA (Sheldon): three visual styles (*Hand-Crafted*, *AI-Reconstructed*, *Non-Stylized*) $\times$ two voice types (*Cloned*, *Synthesized*). Participants' familiarity with Sheldon was measured via questionnaire and used to conduct post-hoc subgroup analyses.

ABSTRACT

Recent advances in generative AI and virtual reality (VR) enable highly realistic celebrity embodied conversational agents (ECAs). However, it remains underexplored how visual stylization and voice type shape users' trust and social presence in celebrity ECA interactions. To address this gap, we examine the effects of visual style (*Hand-Crafted*, *AI-Reconstructed*, *Non-Stylized*) and voice type (*Cloned*, *Synthesized*) by conducting a 3×2 within-subjects study ($N = 36$). Results show that visual style significantly influenced trust and social presence, whereas voice showed no main effect but interacted with style on trust. A post-hoc subgroup analysis based on self-reported familiarity suggests that higher familiarity was associated with higher overall trust and social presence, while lower familiarity corresponded to greater sensitivity to stylistic variation. These findings provide empirical support for extending parasocial interaction frameworks to immersive VR contexts.

Index Terms: Embodied Conversational Agent, Virtual Reality, Visual Stylization, Voice, Virtual Human, Parasocial Interaction

1 INTRODUCTION

Embodied Conversational Agents (ECAs) are increasingly used in counseling, education, and healthcare. Recent advances in AI-driven stylization and voice synthesis have enabled celebrity-based ECAs. Prior work suggests that appearance and voice cues shape trust: Gao et al. [1] studied these effects for non-celebrity agents

in non-conversational settings, and Lupetti et al. [3] examined how ECA appearance and conversational style influence trust in health-care interactions. However, how visual style and vocal cues jointly shape users' trust and social presence in celebrity ECA interaction remains underexplored.

In this study, we examine how AI-driven visual stylization and voice cloning shape users' *Trust* and *Social Presence* in conversations with a celebrity ECA. We built a celebrity ECA based on a well-known TV character (Sheldon) and conducted a within-subjects VR study ($N = 36$) varying visual style and voice type. We additionally conducted a post-hoc subgroup analysis based on participants' self-reported celebrity familiarity to examine whether effects differ by familiarity. Our findings provide empirical support for the applicability of parasocial interaction frameworks in immersive VR contexts.

2 EXPERIMENT

We used a 3×2 within-subjects design with *Style* (*Hand-Crafted*, *AI-Reconstructed*, *Non-Stylized*) and *Voice* (*Cloned*, *Synthesized*). Based on job-interview clips from *The Big Bang Theory* (Season 3, Episode 14), we created Sheldon ECA stimuli for each condition.

For *Hand-Crafted*, we manually modeled Sheldon from a frontal frame of the baseline video to preserve key facial features. For *AI-Reconstructed*, we used the same frame to generate a facial mesh and converted it to a MetaHuman model via a learning-based reconstruction approach [1]. For *Non-Stylized*, we used the MetaHuman preset *Myles* as a neutral baseline without Sheldon-specific traits. Figure 1 shows the three styles. For *Synthesized* voice, we used Microsoft Azure Speech with the preset voice "Guy (en-US)".¹ We

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¹<https://azure.microsoft.com/en-US/products/ai-services/ai-speech/>

Measure	Item	Statistical Method	Effect	F	p
Trust	<i>Capable</i>	ART ANOVA	<i>Style</i>	$F(2.00, 175.00) = 5.585$	0.004**
	<i>Ethical</i>	RM ANOVA	<i>Style</i>	$F(1.95, 68.25) = 4.348$	0.017*
			<i>Style * Voice</i>	$F(1.97, 69.01) = 3.739$	0.029*
	<i>Reliable</i>	RM ANOVA	<i>Style</i>	$F(1.82, 63.73) = 7.956$	0.001**
Social Presence	<i>Social Presence</i>	RM ANOVA	<i>Style</i>	$F(2.00, 175.00) = 8.244$	0.0004***

Table 1: Significant overall main and interaction effects of *Style* and *Voice* on *Trust* and *Social Presence* (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

Familiarity Group	Measure	Item	Statistical Method	Effect	F	p
High Familiarity	<i>Trust</i>	<i>Capable</i>	RM ANOVA	<i>Style</i>	$F(1.66, 29.96) = 4.243$	0.030*
	<i>Social Presence</i>	<i>Social Presence</i>	RM ANOVA	<i>Style</i>	$F(1.41, 25.38) = 6.147$	0.012*
Low Familiarity	<i>Trust</i>	<i>Capable</i>	RM ANOVA	<i>Style</i>	$F(1.83, 29.25) = 4.260$	0.027*
		<i>Ethical</i>	RM ANOVA	<i>Style</i>	$F(1.81, 28.98) = 3.424$	0.050*
				<i>Style * Voice</i>	$F(1.93, 30.80) = 4.880$	0.015*
		<i>Reliable</i>	RM ANOVA	<i>Style</i>	$F(1.84, 29.38) = 8.772$	0.001**
				<i>Style * Voice</i>	$F(1.69, 27.01) = 3.643$	0.047*

Table 2: Post-hoc subgroup analysis by self-reported celebrity familiarity. Participants were divided into *High Familiarity* (rating ≥ 3) and *Low Familiarity* (rating < 3) groups based on a 5-point familiarity scale. We report significant effects only (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

fine-tuned GPT-SoVITS [4] on Sheldon speech extracted from the baseline clips for higher-quality cloning than zero-shot methods. We measured *Trust* with the Multidimensional Measure of Trust (MDMT) [5] (*Capable*, *Ethical*, *Sincere*, *Reliable*) and *Social Presence* with Higgins et al.'s scale [2]; all items used 7-point Likert ratings.

The study was IRB-approved. We recruited 36 participants (21 female, 15 male; $M=25.33$, $SD=3.83$) who rated Sheldon familiarity (5-point) Post-hoc, we formed *High* vs. *Low Familiarity* groups (rating ≥ 3 vs. < 3 ; $n=18$ each). After consent and demographics, participants role-played a female recruiter in a VR interview, responding to Sheldon via on-screen captions and his speech/facial expressions. Interactions followed a Wizard-of-Oz procedure; questionnaires were completed after each condition, with breaks/withdrawal allowed. Participants received 5 minutes of training; sessions lasted 30-40 minutes, with \$7 compensation.

3 EXPERIMENT RESULTS

We analyzed the effects of *Style* and *Voice* on *Trust* and *Social Presence* using two-way repeated-measures ANOVA when assumptions were met, and Aligned Rank Transform (ART) ANOVA otherwise. We further conducted post-hoc subgroup comparisons based on self-reported celebrity familiarity.

Overall Effects of *Style* and *Voice* on *Trust* and *Social Presence*. Table 1 summarizes significant main and interaction effects on *Trust* and *Social Presence*. For *Trust*, *Style* showed significant main effects on *Capable*, *Ethical*, and *Reliable*, whereas *Voice* showed no significant main effects but interacted with *Style* on *Ethical*. No significant effects were observed for *Sincere*. *Style* also significantly affected *Social Presence*, while *Voice* showed no significant effects.

Subgroup Effects of *Style* and *Voice* by Familiarity. Table 2 summarizes significant effects on *Trust* and *Social Presence* across familiarity subgroups. For the *High Familiarity* group, *Style* showed a significant main effect on *Capable* trust and *Social Presence*, while *Voice* exhibited no significant main or interaction effects; pairwise comparisons among the three styles were not significant. In contrast, for the *Low Familiarity* group, *Style* significantly affected *Capable*, *Ethical*, and *Reliable*, with *Style*×*Voice* interactions on *Ethical* and *Reliable*; no significant effects were found for *Social Presence*. Overall, higher self-reported familiarity was associated with higher trust and social presence and lower sensitivity

to style/voice variation, whereas lower familiarity showed greater cue sensitivity, suggesting that unfamiliar users may require tighter style–voice alignment.

4 CONCLUSION AND FUTURE WORK

We investigated how visual style and voice type shape users' *Trust* and *Social Presence* when interacting with a celebrity ECA in immersive VR. Using clips from *The Big Bang Theory*, we built Sheldon ECAs and conducted a 3×2 within-subjects study. Results show that *Style* significantly affected both *Trust* and *Social Presence*, whereas *Voice* showed no main effect but interacted with *Style* on *Trust*. Post-hoc subgroup analyses further suggested that self-reported familiarity with Sheldon was associated with differences in these effects, with lower-familiarity participants more sensitive to stylistic variation. Overall, our findings offer design guidance for trustworthy celebrity ECAs with generative AI and empirically extend parasocial interaction theories to immersive VR.

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